

Go-Go Biome: Evaluation of a Casual Game for Gut Health Engagement and Reflection

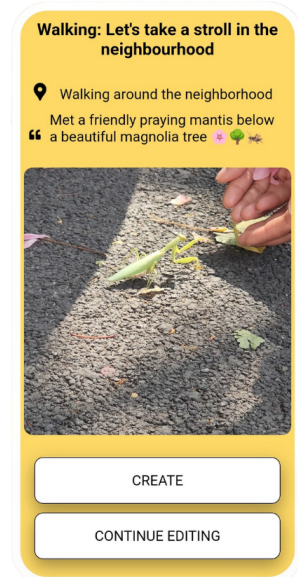
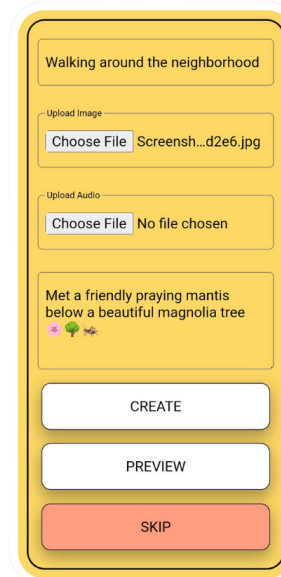
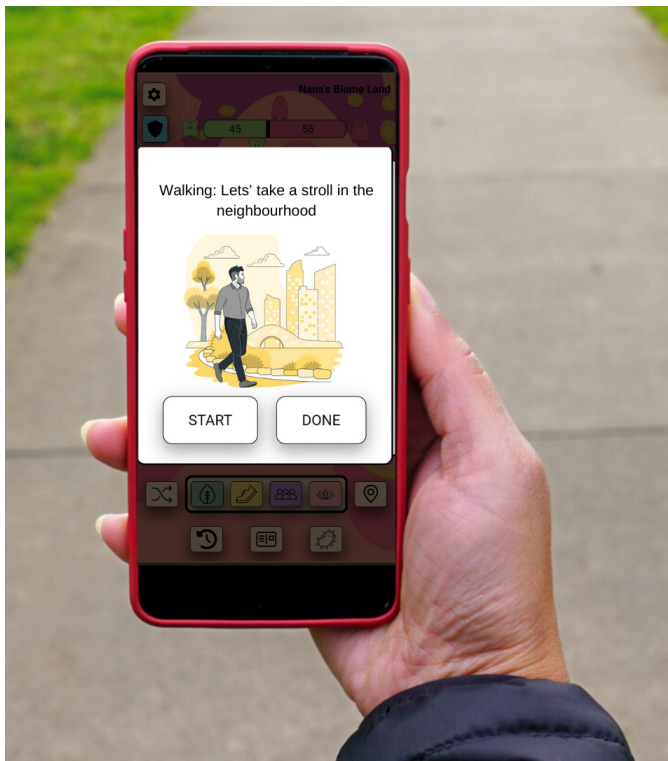
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GO-GO BIOME

Figure 1: (Left) Go-Go Biome in action with a player engaging in a physical activity that involves taking a stroll in the neighbourhood. (Right) A template to log activity experiences opens right after the player ends the activity.

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ABSTRACT

Experts emphasise that maintaining a healthy gut microbial balance requires the public to understand factors beyond diet, such as physical activity, lifestyle, and other real-world influences. Games as experiential systems are known to foster playful engagement and reflection. We propose a novel approach to promoting activity engagement for gut health and its reflection through the design of the Go-Go Biome game. The game simulates the interplay between

friendly and unfriendly gut microbes, encouraging real-world activity engagement for gut-microbial balance through interactive visuals, unstructured play mechanics, and reflective design principles. A field study with 14 participants revealed that important facets of our game design led to awareness, playful visualisation, and reflection on factors influencing gut health. Our findings suggest four design lenses—*bio-temporality*, *visceral conversations*, *wellness comparison*, and *inner discovery*, to aid future playful design explorations to foster gut health engagement and reflection.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; Interactive systems and tools;

KEYWORDS

casual game, reflection, gut health, gut microbiome, activity engagement, health reflection

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1 INTRODUCTION

A balanced gut microbiome supports immune health and may aid in preventing autoimmune diseases [16, 103]. However, the rise in sedentary lifestyles and poor diets has heightened gut health issues, underlining the importance of promoting healthy behaviours and awareness towards gut health [43]. Interest surrounding gut health has been mounting in recent years, with several health fads and diet plans making fleeting headlines [108]. While diet is widely recognised for its role in maintaining gut health, the general public's awareness of other factors influencing gut health, like physical activity, lifestyle, engaging with nature, mindfulness, social interaction, and emotions [22, 88], is low [105]. Research indicates that engaging in various activities related to these factors can positively impact the gut microbiome and overall health [22, 88].

A balanced gut microbiome resulting from these engagements may enhance digestion, nutrient absorption, and immune function [90, 99] promoting overall well-being and reducing the risk of gastrointestinal disorders. Additionally, improved gut health is linked to better mood regulation, cognitive function, and weight management, fostering individual mental and physical vitality [88]. Encouraging individuals to reflect on their everyday habits and activities that can influence gut health may play a pivotal role in discerning patterns that contribute to imbalances in the gut microbiome, encouraging the adoption of new or alternative habits that positively impact this balance. However, the microscopic nature of gut microbes presents a challenge in thinking about their role in gut health, presenting a need for playful visualisation of these intricate inner processes [77].

This paper aims to establish a holistic approach to gut health through the design and evaluation of a casual smartphone game called Go-Go Biome (GGB) (Figure 1) for gut health engagement

and reflection. Unlike traditional video games, casual games have simple rules, are easy to learn, and require minimal expertise [54]. They focus on real-world concepts, offering purpose beyond mere entertainment. Smartphone games can allow for pervasive play and glanceable real-time feedback through their ubiquitous presence that supports visual cues, notifications, and quick play sessions [29, 84]. Such features embedded within a casual game for gut health engagement can help players make gut-friendly decisions in everyday life. Drawing on these benefits, we employed research through design (RtD) [109] and game design methodology [37] to design Go-Go Biome (GGB), a casual smartphone game rooted in the concept of unstructured play [40, 101], affording players greater flexibility to engage in real-world activities through self-directed play (ibid). Unlike conventional sensing technologies, this game relies on manual input from players to progress in the game. GGB takes an alternative approach to designing engaging play experiences by drawing inspiration from the inner workings of the gut, such as the gut microbiome, intestines, and external factors that influence gut health. Through the design of GGB, we aimed to provide non-human entities, such as gut bacteria, with a voice to help players develop a deeper understanding of their needs and their significance in maintaining gut health.

The research questions for this study aimed to examine: 1) participants' general understanding of the topic of gut health; 2) how unstructured play directed by a casual smartphone game supports activity engagement for gut health; 3) the role of the designed features in enabling reflection on personal values, choices, and behaviours that may affect gut microbiome and health. We conducted a field study of GGB with 14 participants over two weeks. Through pre- and post-study interviews and player logs, we studied participants' interaction with the game, game activities, and their role in supporting gut health reflection.

Reflection, a widely discussed topic, has various conceptual definitions stemming from a variety of fields including psychology, philosophy, and education [92]. Theoretical perspectives on reflection range from Dewey's systematic approach to problem-solving [4, 25], Moon's transformative re-evaluation for conceptual change [69], to Schön's practical 'reflection-in-action' and 'reflection-on-action' for enhancing professional practice [92, 96]. A recent synthesis of these perspectives on reflection as offered by Mols et al. defines reflection as the analysis of past, present, and future experiences to refine thoughts, beliefs, and actions in daily life [68]. This comprehensive definition that encapsulates and extends upon earlier perspectives, offers a foundational understanding that is central to this paper's exploration of reflection.

Additionally, the following works on designing for reflection in HCI influenced the design and analysis of GGB's mechanics: 1. reflective design strategies by Sengers et al. [93], 2. reflective game design principles by Khaled [45], and 3. the taxonomy of design resources for reflection by Bentvelzen [8]. By combining these design strategies and resources for reflection to frame our results, we articulate which of the designed features supported players in thinking about personal activities that affect gut health with corresponding design implications. Our results guide us to key themes and the development of four design lenses for health engagement and reflection: *bio-temporality*, *visceral conversations*, *wellness comparison*, and *inner discovery*, which specifically extend

the taxonomy of design resources for reflection by Bentvelzen et al. [8].

The contributions made through this work include 1) A system contribution by presenting the design of a casual game for gut health activity engagement and reflection that may inspire future design researchers to develop interactive technologies using unstructured play to facilitate health engagement and reflection; 2) An empirical contribution by presenting insights from our field study through 3 themes can be instrumental in supporting user experience and game design researchers in understanding the potentials and the limitations of using our approach; 3) Theoretical and practical contributions through four design lenses and supporting design considerations to aid the design of games and interactive applications for health reflection.

To contextualise our research, we establish our positionality by presenting the distinct perspectives we bring to it. This research is influenced by a team of multidisciplinary researchers at various professional levels from fields including interaction and user experience design in Human-Computer Interaction (HCI) and Human-Food Interaction (HFI), game design and development, sociology, and gastrointestinal health. Our team is diverse in race, native language, and nationality. The two major paradigmatic approaches influencing our work include ontological and phenomenological assumptions surrounding gut health. The primary researchers study physical and digital interfaces within the context of HCI and HFI centred around play, health, and reflection, practising qualitative approaches to understanding design and user experiences, which we bring to this research. One researcher is a subject matter expert in gastrointestinal health and nutrition, instrumental in examining the application of scientific facts to game mechanics. Four team members have previously designed games and play for health engagement and reflection. Two members have experience with software programming, and one of them has game development experience involving small-scale video games. Next, we dive into the wide range of literature that influenced our research.

2 RELATED WORK

We start this section with a brief overview of current approaches used for gut health understanding, followed by games, and gamified systems for healthy activity engagement and visualisation, where reflection was either part of their design goal or an outcome in their study findings.

2.1 Current approaches for gut health understanding

Current information sources that address gut health and topics such as the microbiome include web articles [1], scientific publications [16], and mobile applications [13, 18]. Besides these passive forms of knowledge, there are interactive games based on structured play, such as board games [79] and digital games [44]. For example, Pasumathy et al. designed a board game to engage people on the broader factors that affect gut health through the physical rule-based representation of information. Their study revealed that the game served as an open forum for meaningful social discussion between players on gut health while learning about the critical real-world factors that affect it. However, the information gained

through a board game may only remain at an awareness level. It may not translate to lived experiences involving real-world activity engagement for gut health, which we aim to address. Phenomenological approaches recognise bodily engagement as key to reflective attentiveness in lived experiences [104]. Engaging players in real-world activities within gameplay can foster reflection on their experiences and the contexts influencing gut health. This approach aligns with Viktor Shlovsky's concept of estrangement and defamiliarisation, making the ordinary extraordinary [64]. Along with engagement, supporting people with playful visualisation of their inner gut process can aid in understanding cause-effect relationships between the broader external factors and internal players, such as the gut microbiome, which we explore through our work.

2.2 Games and gamified systems for health activity engagement

The lack of games and interactive systems within HCI research that explicitly address gut health engagement and reflection led us to broaden our search to games and gamified applications and the features popularly used to encourage health-related activity engagement. Previous works in HCI promoting health-related activity engagement [30, 80, 86] include digital games [75, 82], gamified health applications for mobile devices [3, 81, 97, 102], physical games [78], and hybrid games [106]. Popular techniques and features used within these technologies to motivate and support activity engagement include playful visualisation [48, 71, 79], storytelling [91], sensing technologies [6, 58], reflective informatics and activity data including ambient systems [34, 60, 71, 106], narrative-based interactions [12, 62], synchronous and asynchronous feedback [34, 48, 53], photo journaling [10, 31]. Our research builds on some of these features in the design for health activity engagement and reflection.

HCI research increasingly focuses on technology-driven tools that leverage goal-setting mechanisms and gamified design components to promote health activity engagement. Previous works have used various game design concepts regarding direct and indirect goal setting, including challenges, quests, virtual rewards, streaks, and competition, to motivate player engagement and help them achieve health goals [9, 21, 24, 59]. Novel goal-setting mechanics include the option of setting both primary and secondary goals for activity engagement see GoalPost [70], choosing from a variety of activity goals, see Bounce [61], and setting one's own goals instead of system-assigned goals, see UbiFit [21]. Assessments of these instruments indicated that basic gamification strategies, such as badge systems, might possess restricted motivational efficacy for engagement [70], and competitive elements could potentially deter motivation [59, 107]. These studies reveal that while goal-setting assistance is vital, users value autonomy in shaping goals that align with their capacities.

Furthermore, engaging users in manually logging their activity data through journaling activities and providing an absolute reference for comparison of user activity level can lead to moments of self-reflection on actions taken towards their health and motivate further engagement. Such logging can lead to moments of self-reflection and open pathways for re-evaluation of their daily habits

and practices that affect health [61]. Such re-evaluation can be supported by presenting activity snapshots and interactive summaries to users [10, 27, 38].

2.3 Visualisation of health concepts and activity data in games and gamified systems

In studies related to gamified health systems, game design principles have been applied to boost motivation, encourage reflection, employ playful visualisations, and make concepts relatable [24]. Examples include systems like Fish'n Steps and UbiFit Garden, which use metaphorical visualizations to represent physical activity data [21, 59]. MindTerior is a mental health game that measures and visualises players' emotional states through a mindfulness score and changing background visuals [56]. In conveying health concepts, Monster Appetite [38], a nutrition-focused mobile game, uses a combination of positive and negative messages and contrasting visuals to influence users' food choices, leading to improved snack selections in the presence of positive messages alongside negative visuals. Collectively, these studies highlight gaps and offer design suggestions. Firstly, visualisations in health activity engagement can improve relatability to health concepts, boosting user engagement. They also show the potential of using positive and negative health visualisations and metaphors to support health activity engagement and reflection. However, the lack of playful visualisation and reflection tools for gut health necessitates an approach that can help people visually connect broader factors to the gut microbiome. While physical, environmental, social, and mindfulness activities are known to support overall health, their role in gut health isn't widely understood. Promoting this understanding can provide a new perspective on how these everyday activities are viewed and understood in the context of gut health. To address this, we looked into the commonly used design strategies and resources for reflection within HCI.

2.4 Reflection in HCI

Reflection as a core element in designing interactive technologies is increasingly popular in HCI [8, 23, 26, 32, 48, 67, 94]. Prior HCI research suggests that reflection cannot be just an outcome of the design, instead it needs to be thoughtfully intertwined into the design of the artefact or experience [45, 65, 93]. This led to the development of a variety of design strategies, considerations, and resources to support design for reflection [8, 45, 93]. For example, the design considerations and strategies put forth by Khaled [45] and Sengers [93] act as thinking tools to guide the design process, but they do not offer insight into its practical application in the design. In contrast, Bentvelzen et al.'s taxonomy of design resources for reflection [8] offers more explicit directives on how those strategies can be applied using the design resources and patterns in the taxonomy. For this reason, we mainly focus on Bentvelzen's articulation of the four design resources: 1) Temporal perspective, which uses aspects of time such as past [46, 47, 87], future, memories [2, 66], slowness or time-based mechanics [48]; 2) Conversation perspective which relies on conversations with technology [30, 63, 66] and with others [100] as the basis to support reflection; 3) Comparison perspective which is concerned with the reference of the current state to an ideal state as a means to support reflection, that uses two approaches

namely an absolute reference [42] and a social reference [100]; and 4) Discovery perspective which is concerned with offering users an alternative lens or a new perspective to view a situation to prompt reflection. The design patterns include ambiguity Kellogg's Gut Bacteria Reef on Steam, provocation [73], and reframing [35, 49–51]. Overall, these four design perspectives guide the thinking behind the design of our game mechanics. Now although these perspectives have been used across a variety of topics within and outside HCI, their application within the context of gut health remains underexplored. Moreover, many of these works do not articulate how these design resources have been applied in crafting their design features for reflection, which this paper addresses. In the following sections, we highlight the implementation mechanisms of these design resources for reflection within the features of Go-Go Biome.

3 GO-GO BIOME

Go-Go Biome (Figure 2) promotes daily engagement in diverse activities to maintain gut health and microbial balance. The game objective is to balance Buggy, the friendly biome, and Minion, the unfriendly biome's population, to a ratio of 85:15 daily through engagement in diverse real-world activities. Inspired by prior work [79], GGB drives the concept of homeostasis or balance in the gut through the game objective. The game contains four activity card decks (Figure 3): *Go Green* (environment), *Get Active* (physical exertion), *Get Social* (social) and *Go Zen* (mindfulness), each with different activities. These activities increase the friendly biome population, affecting the unfriendly population inversely, as seen on the Biome Population Metre. The diversity of activities enhances the visual vibrancy of the biome landscape on the home screen. The metre resets each night for a new start each day, aligned with the ideology of taking consistent daily actions for maintaining gut health [105].

Players' daily activities generate diverse microbial characters in the Biome Garden, reflecting the diversity of the previous day's activity. However, inactivity for two or more days gradually reduces the friendly biomes in the garden. The Biome Energy Status on the game's home screen (Figure 2- top left) shows the player's friendly biome's energy level, encouraging them to undertake activities to boost it. Players respond to these prompts by selecting an activity card from any deck and engaging in it. The player manually logs activities. Upon activity completion, players can also document their activity experiences, which are saved as logs under the My Cards tab. Engaging in diverse activities helps the player's friendly biome character gain energy in the form of steps to progress on the Gut Trail map. As players advance on this trail they defeat unfriendly biomes along the way and maintain their trail. This process forms an endless loop as players continually strive to balance their friendly and unfriendly biome populations daily through their activities. A scenario of the gameplay in action is presented in Figure 4.

3.1 Designing Go-Go Biome

This section briefly explains GGB's core functionalities, their role in the game, guiding theories, and our design reasoning (Table 1). This paper focuses on GGB's features and its evaluation through a field study aimed at understanding its role in supporting real-world

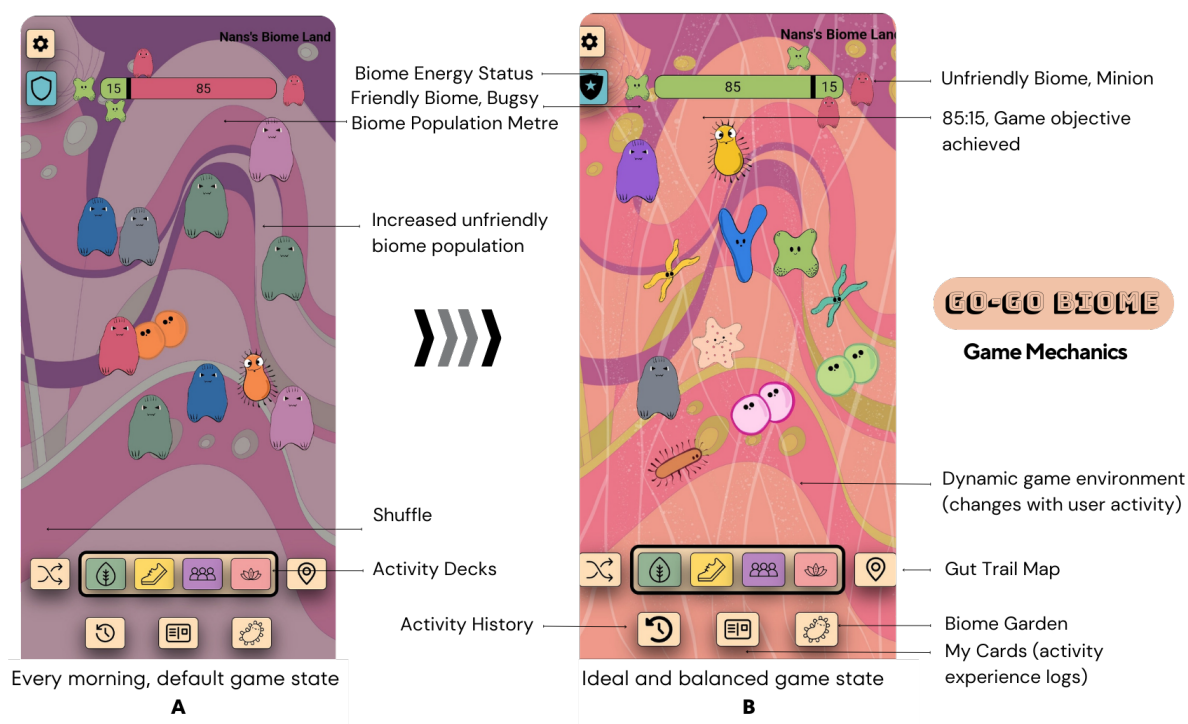


Figure 2: Go-Go Biome’s primary game features and a view of the game’s home screen. (Left to Right) Shows the transition of the game’s state (biome land) with player engagement in diverse activities across a single day.



Figure 3: The game offers variation in the play activities through four card decks, namely Go Green, Get Active, Get Social and Go Zen. The player chooses an activity from these decks to play the game.

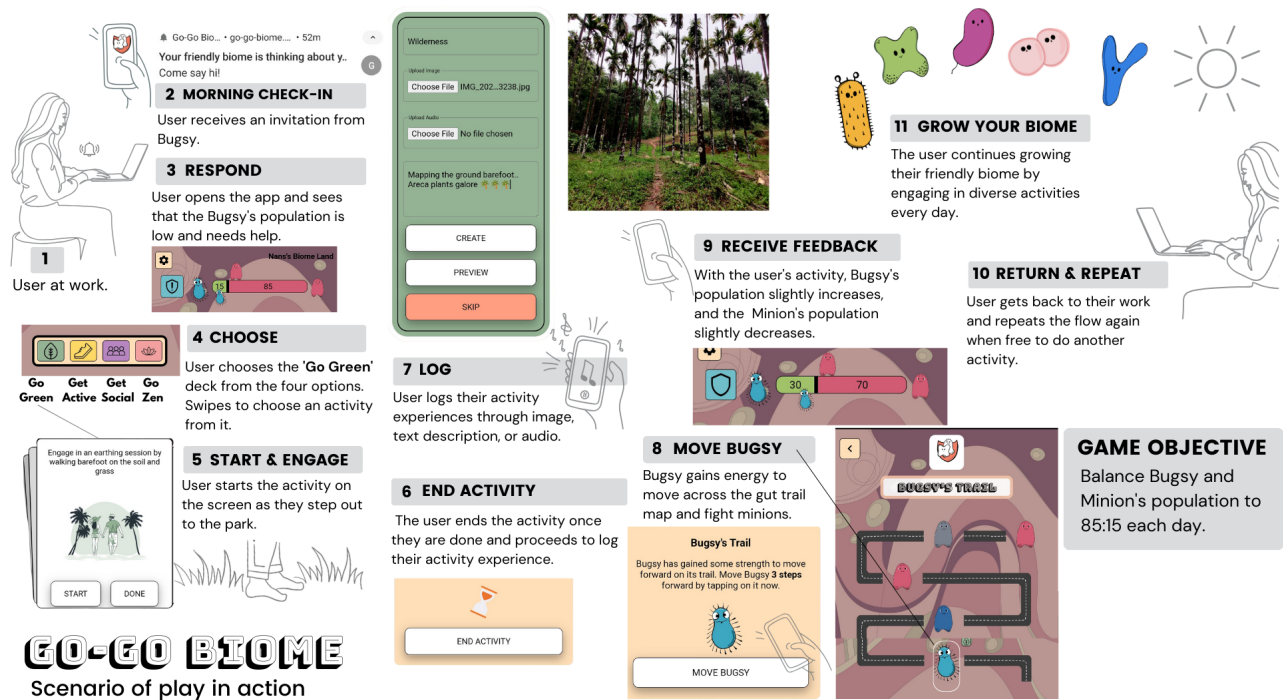


Figure 4: Go-Go Biome, a scenario of play in action involving player interaction with the game and engagement in a real-world activity to help the friendly biome, Buggy in the game.

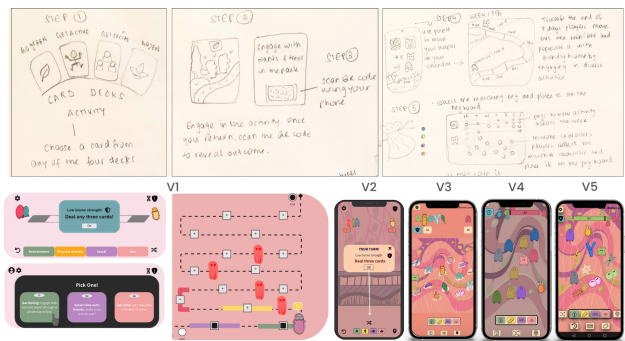


Figure 5: Initial sketches (top) and the evolution of Go-Go Biome's design through paper prototypes and wireframes into a high-fidelity application (bottom right).

activity engagement for gut health, and in fostering participant reflection on everyday habits that may affect it. Thus, this paper mainly examines participants' experiences with the reflective design strategies and resources integrated into the game's features and implications.

The initial design phase of the game included character creation, narrative development, storyboarding, prototyping, and multi-type testing, such as on paper and wireframes. Figure 5 shows early versions crucial to defining key features and the iterations involved. The game progressed through an iterative process over five versions, refining its narrative, improving interface aesthetics and functionality, and resolving usability issues. This paper presents the final version (V5), evaluated in a field study. The table below (Table

1) outlines GGB's key game features, their role, design resources employed, and rationale.

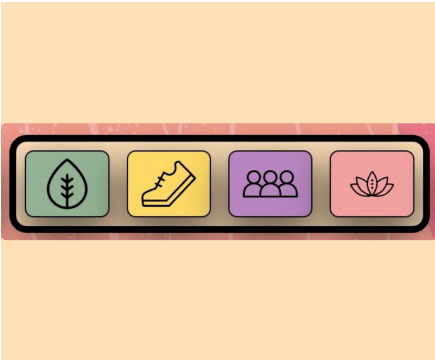
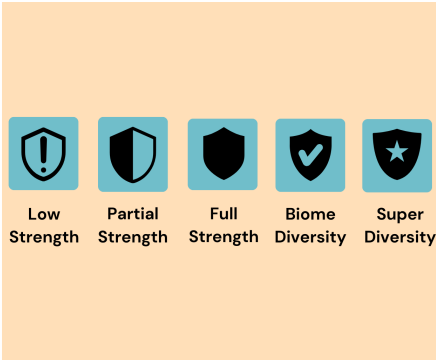
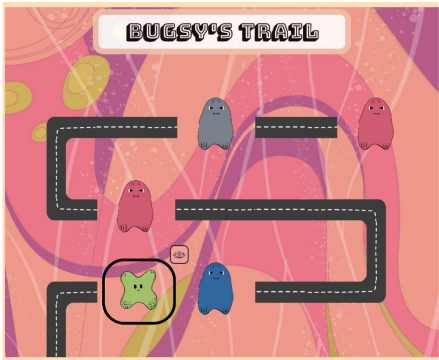
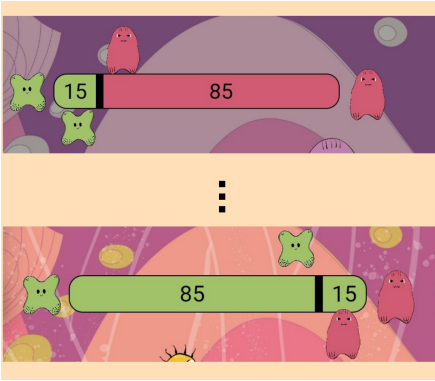
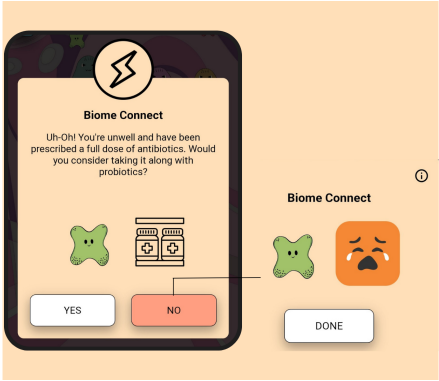
4 USER STUDY: GO-GO BIOME

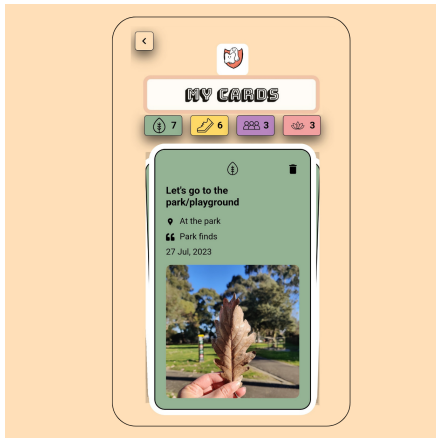
Go-Go Biome was tested in a 2-week field study to capture, understand, and evaluate participant interactions with the game in everyday life involving real-world contexts. Furthermore, we sought to investigate how the game influenced participants' attitudes and behaviours related to their health activities and their overall experience with the game features and feedback. The University Ethics Board approved the study. The study inclusion criteria involved adults aged 18 years and above who were interested in gut health and willing to engage in activities directed through game interactions. We shared the recruitment flyer on Facebook game groups and social media platforms of the primary and secondary circles of the research team within and outside the University. We went for convenience sampling using snowballing techniques from the interested pool of participants. We did not offer financial compensation to participants.

The total recruited participants included 14 (6 Males, 8 Females) between 25 and 37 years of age from diverse ethnic backgrounds (Australian, Bangladeshi, Colombian, and Indian) residing in an urban metropolitan city. Participants were requested to play the game for at least 6 days, and up to 12 days at most, requiring a minimum of 30 minutes of daily activity engagement.

Balancing the biome population to 85:15, the main game objective, demanded participants to engage in a minimum of 30 minutes of daily activity, typically involving 3-5 activities. This level of commitment resulted in a limited sample size and a brief study duration.

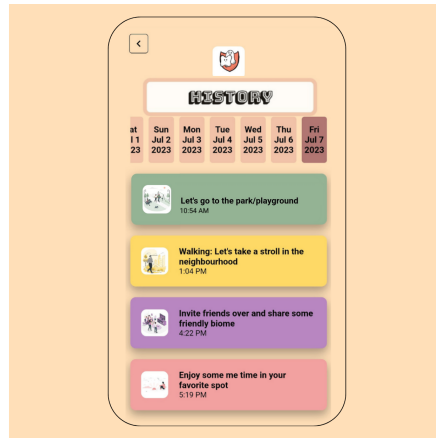
Table 1: List of Go-Go Biome’s features and their use, including associated design concepts and rationale.

		
Activity Decks (Go Green, Get Active, Get Social, Go Zen): Wide offering of game activities. Design Resources and Reasoning: Unstructured play mechanics: To provide player autonomy to self-direct play, giving the license to participate [93]; Reframing familiar activities in gut health context (ibid). Activities were informed by scientific literature [22, 36, 41, 88] and an expert. Arbitrary points (between 10 and 30) are assigned to each activity based on their importance, contributing to the game objective of balancing the friendly and unfriendly biome population to 85:15.	Biome Energy Status: Depicts the energy of the friendly biome ‘Bugsy’ that changes across five levels with player activity and in-activity. Design Resources and Reasoning: Slowness and the present: To enable relatability to the biome’s energy level and its slow transition with player activity [89].	Gut Trail Game Map: Offers a sense of progression after each activity depicted through the movement of the friendly biome on an endless trail. Design Resources and Reasoning: Unstructured play mechanics: Give players the license to participate [93]; Representative of the human intestine to reinforce that gut health is not a destination but a journey requiring small and consistent action.
		
Biome Population Metre: Depicts balance between the player’s friendly and unfriendly biome population that changes with player activity. Design Resources and Reasoning: Comparison, discovery, ambiguity: Designing for interpretive flexibility [93]; Provide dynamic feedback to players (ibid). To allow a comparison of the present state to an ideal state, [74] conveys the concept of biome balance and ‘homeostasis’ as depicted by the game’s objective of 85:15.	Biome Garden: Houses the growth of friendly biome characters that show up days after player activity. Biome Emoji: Depicts the emotional state of the player’s friendly biome. Design Resources and Reasoning: Slowness and the future: Anticipation of the future, cause-effect, fostering the attentive practice of noticing [19], abstract representation of inner processes through the slow growth of the friendly biome characters and its humanisation [89, 95].	Biome Connect: Asks yes or no, what-if questions twice a day. Players respond with a yes or no, receiving an emoji response to their answer from their friendly biome. Design Resources and Reasoning: Dialogic engagement between players and the friendly biome mediated by technology to support in-situ or momentary reflection [63, 66, 93]; Provocation and ambiguity [26]; critical self-observation [45]: Enabling thinking beyond the self to consider subtler aspects such as the gut microbiome and the daily lifestyle choices that affect it.



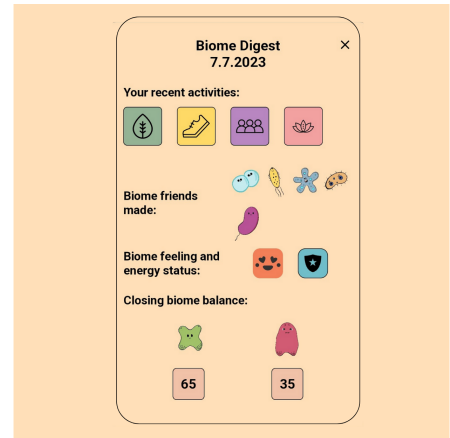
My Cards (Activity Experience Logs): To capture activity experiences through image, text input, and audio recordings.

Design Resources and Reasoning: Memories [94], designing for self-expression, inspiring rich feedback from players, serving as memories that support reflection on their lives [93], and Encouraging agency and ownership over self-directed actions taken towards gut health [7].



Activity History: Provides a calendared display of the player's logged activities.

Design Resources and Reasoning: Past and the future; Design to spot patterns in activity and its diversity; re-assess past activity and plan for the future [8].



Biome Digest: A snapshot of the previous day's activity, diversity, the level of biome population balance achieved, and friendly biome earned or lost.

Design Resources and Reasoning: Past and the future: Momentary reflection on the previous day's activity to inspire action [94].

Despite this shortcoming, their consistent engagement (see Figure 6) with Go-Go Biome allowed us to gather rich and crucial data. The study duration and sample are consistent with prior CHI papers with similar exploratory goals [17].

Eleven out of 14 participants had low to no knowledge of gut health; three participants had a low-level conceptual understanding of a few factors that affect gut health. Their level of education ranged from undergraduate to doctoral level of study. The pool involved a mix of students and working professionals. All participants were curious to learn about factors affecting gut health and were excited to engage with the topic through play. Participants played the game at their convenience, at different locations based on their comfort and preference. Participants were asked to interact with the game and engage in the game activities at least one or more times every day.

Our approach aligns with influential work in developing mobile-based tools for physical activity engagement [20, 21, 70, 71] emphasising the collection of natural, qualitative information in participants' day-to-day lives. It's important to note that the primary goal of this study was not to demonstrate behaviour change conclusively, which may not be feasible or appropriate in the context and developmental stage of HCI research [52]. A longitudinal study would be necessary to assess the system's sustained effects.

Instead, we focused on gathering in-situ data to assess the practical viability of interacting with GGB in real-world settings. By considering GGB as a research probe, we also aimed to better understand players' interaction with and experience using reflection-based design features embedded in the system. Furthermore, to ascertain their role in enabling reflection on personal values, practices, and actions surrounding gut health.

4.1 Study stages

The user study involved three stages: (1) The pre-study interview questioned participants on their understanding of gut health and their current habits and practices surrounding diet and lifestyle factors, including weekly routines, health values, and priorities. (2) The field study deployed and evaluated GGB in real-world contexts. Participants interacted with the game daily, engaging in one or more real-world activities from the game decks. (3) The post-study interview questioned participants on their interactions and experiences with the various game features and their role in supporting reflection on gut health factors, including personal habits that influence them. The primary author debriefed participants in the pre- and post-study interview stages on gut health factors introduced in the game, gathering their understanding of it.

4.2 Data collection and analysis

Data was collected in all three stages of the study. Qualitative data was collected through audio and video from interviews, images, text descriptions, and audio recordings from the game application. The primary researcher loosely referred to an interview guide during the interviews to cover all the features and key concepts designed. However, discussions were typically guided by participant responses based on what was meaningful to them, with the interview topics woven into it. Interviews were semi-structured for further emergence of new topics [11], lasting approximately 45-60 minutes each, and transcribed verbatim.

The pre-study interviews questioned participants on the general understanding of and assumptions surrounding gut health, including perceptions of their gut health and well-being, values, practices, weekly rituals, and their current reliance on technology to support

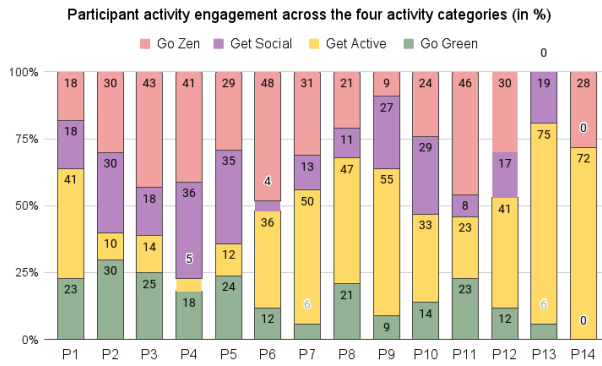


Figure 6: Participant activity engagement across the four activity categories.

health activity engagement. Participants were on-boarded virtually and provided a conceptual brief regarding the game concept before the study started. The study involved participant engagement in various game activities based on their preferences, and they were instructed to engage with the game daily. The post-study interviews questioned participants on their experiences from interacting with GGB’s features, its role in supporting playful engagement in health activities and aiding reflection on their health choices, behaviours, and values. Additionally, game analytics was collected from the app’s back-end to understand the frequency of participant interactions with the game decks, choice of activities, activity diversity, etc., presented as descriptive statistics (Figure 6).

We conducted reflexive thematic Analysis (TA) given the ontological, theoretical, and paradigmatic underpinnings influencing data collection and analysis [14]. This approach helps to effectively capture participants’ subjective experiences and narratives, simultaneously recognising the reflexive impact of the researcher’s interpretations (ibid). The first author, who conducted the thematic analysis, has previously designed games for health engagement and reflection. The fifth author who participated in forming design themes has worked in designing play-based technologies for reflection. Therefore, this analysis is influenced by knowledge of games and play-based technologies for health reflection. Data was thematically coded and analysed by the first author using the six steps to TA [14] involving both inductive and deductive approaches [15].

Initial coding was inductive, with codes describing participant experiences of interacting with different features and their interpretations of the game concept; the effects of game features on topic understanding and in enabling closer connections to the concept of the gut microbiome; general feedback on the app’s usability, challenges, and suggestions; and features that led to real-world changes in participant mindsets, values, and actions surrounding gut health. Through this, we learned that the game was generally successful in translating the topic of gut health and motivating participant engagement in the game activities. As the analysis progressed, we found ourselves leaning towards a deductive approach, understanding the data through the lens of reflection [8] and learning how the designed features affected participant reflection on personal health values, practices, and actions. Collectively, this led us to 11 recurring

topics (not themes)- “challenges”, intentions”, “priorities”, “context”, “mood”, “commitment”, “variety”, “empowerment”, “visualisation”, “progression”, and “relatability”. Each of these topics contained sub-codes that were further grouped and regrouped during the selective coding phase, eventually generating six themes. The themes were discussed with the fifth author and collaboratively iterated to form three high-level themes surrounding “awareness”, “embodiment”, and “ownership” surrounding gut health. We employed interpretivist and constructionist paradigms that acknowledge the meaning and meaningfulness of participant experiences and recurrence [14]. The influence of our approach is mirrored in the way the themes and findings are organised and how they relate to one another. Overall, the data generated included 864 data sets (one data set = 1Q + 1A). Recurring patterns were iteratively identified, leading to 56 codes and three high-level themes.

5 FINDINGS

We report the insights from our field study across three themes: Incremental Awareness, Embodied Connections, and Health Ownership. Overall, we found that participants played the game for an average of 9 days (minimum 6; maximum 14 days). The average number of activities engaged during the study duration is 18 (minimum 7; maximum 28) (Figure 6). The average number of daily activities is 2 (minimum 1; maximum 3). We present our findings (F) below as F1, F2, F3, etc.

5.1 Theme 1: Incremental Awareness

This theme illustrates how GGB shifted participants’ perception of gut health from an abstract idea to a tangible one, offering a fresh view of routine activities and promoting mindful decision-making for healthier lifestyles. The game’s features stimulated reflection on everyday routines, prompting players to identify and alter habits harmful to gut health. The game advocated for micro-changes in health behaviours by emphasising the substantial influence of small lifestyle modifications on gut microbial diversity and overall health.

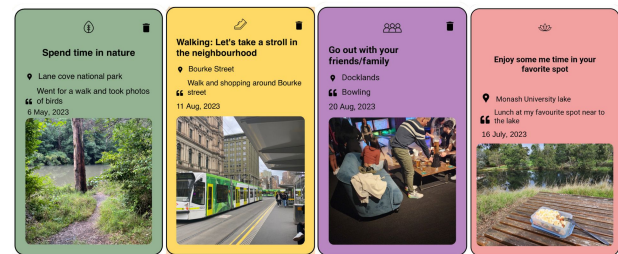


Figure 7: Screenshots of participant activity experience logs involving nature interaction, physical exertion, social, and mindfulness activities.

5.1.1 F1: Go-Go Biome altered understanding of gut health. Pre-study interviews revealed that gut health is challenging to visualise and reflect on. P6 said, “We think of our gut health only when something goes wrong, but not when you feel good.” However, interacting with the game concepts helped them visualise and relate to the connections between everyday activities and gut health. P6 (continued), “The app’s visuals offered a relatable representation, making it easier

to reflect on the daily impact of our activities on gut health, which otherwise might be challenging." The understanding they gained added more meaning to their routine activities. P8 added, "The game helped me understand how even the routine things I do can have a meaningful impact."

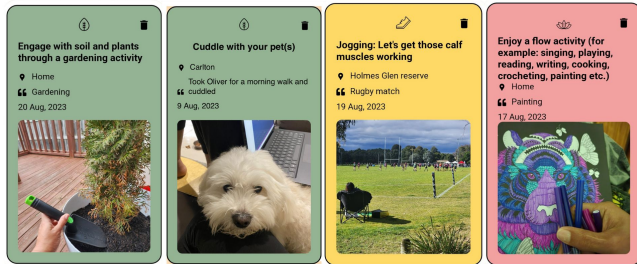


Figure 8: Screenshots of participant activity experience logs from the study capturing multiple activity contexts.

Furthermore, it widened their perspective on how they could care for their gut health (Figure 6). P13, "I learnt that it does not always have to be exerting in the gym. Instead, one could consider a few simple activities." Overall, it helped to reframe their everyday activities, viewing them in a new light. P6 expressed, "My mindset about these activities has changed as these are no longer just activities I do during my break or between work. Rather, they help to maintain my gut health."

5.1.2 F2: Contextual awareness of gut health factors enabled mindful decision-making. Participation in the game activities heightened participants' awareness of broader factors impacting gut health, like physical contexts, including their location and environment, thereby promoting the outdoors. For example, P6 said, "I enjoyed playing the card 'Enjoy some me time in your favourite spot.' It made me aware of my surroundings and encouraged me to get outside and find a spot to enjoy during my break." (Figure 7) Extension of their awareness beyond the game world and into their everyday life was suggested by (P2), who said, "At times, in specific places like the mall, at the park, or while playing with my pet, I wondered about the links between these activities and what it could do for my gut microbiome." P4 said, "For example, whenever I take a walk, or even when I'm stressed, I think of how it may affect them (internal biome)." Overall, participants understood the importance of activity diversity to maintain good gut health. P10 said, "I understand now that engaging in a wider set of activities than the same ones can help better maintain your gut health as they have a direct and indirect impact on it." Suggesting the willingness to make positive changes, P10 said, "(...) I automatically want to include more beneficial activities rather than sliding back."

5.1.3 F3: Game features helped to reassess and compensate for missed activities. Interacting with the Biome Digest every morning and viewing their activity history allowed participants to compare their present and past activities, helping them reassess and plan their future activities. P2 expressed, "I enjoyed receiving the Biome Digest every morning as it presented me with my previous day's activity, making me think of what I could do during the day." Viewing

past activities invited participants to reflect and act on missed opportunities. P13 said, "Going through my activity history (...) made me wonder why I didn't engage in any activity from the Zen deck and to see if I can include some in my routine." Similarly, logging daily activities helped participants reflect on activities missed for the day (Figure 10). P3 reflected, "If I haven't done any physical activity the whole day, I would think about it when I log my other activities." Revisiting past activities helped to recognise monotony in one's life and compensate for missed activities. P11 said, "Looking at my Activity History, I realised the monotony in my routine. Having a record was good to see if I did any other activities to compensate for the days I did not go to the gym."

5.1.4 F4: Game engagement supported self-recognition of habits affecting gut health. Post-study interviews revealed self-awareness in participants as they reported on personal habits that may negatively influence their gut health and microbiome. Regarding the lack of movement, P4 said, "I now recognise that sitting for long hours at my desk and not moving or engaging in activities outside of work may contribute to the increase of unfriendly biome and discourage the growth of friendly gut biome." Regarding unhealthy lifestyle choices, P13 said, "Smoking and eating junk food are two habits that may be affecting my gut microbes and health." Regarding lack of meal diversity, P12 said, "Not consuming fibre-rich foods, especially vegetables, may be affecting my gut health."

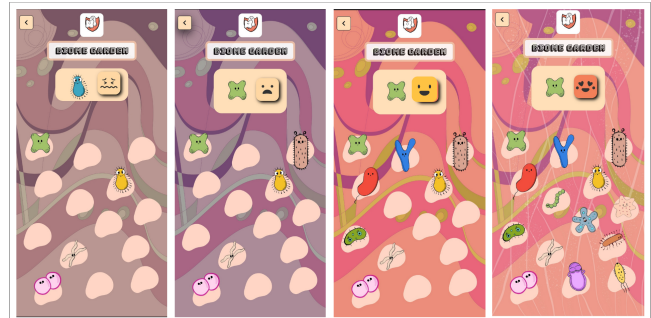


Figure 9: Asynchronous feedback that shows progression through the slow growth of the friendly biome in the biome garden across five days of activity. The image also depicts the emotional state of the friendly biome, Bugsy, involving emojis and sound.

5.1.5 F5: Go-Go Biome fostered micro-changes in health behaviours and mindset. GGB encouraged participants to incorporate micro-changes into their routine, swapping old habits for new healthy behaviours. P11 said, "This led me to schedule an appointment with a dietitian to ensure I have more balanced meals throughout the day." Regarding micro-changes, P10 expressed, "I have the habit of snacking once I get back home from work, but I found that my interaction with the game was priming me to go for slightly healthier snacks such as nuts." Regarding replacing old habits (Figure 8), P6 said, "When I have some free time, instead of scrolling through my mobile and watching a video, I now take 10 minutes to engage in one of the activities." Participants believed that making incremental changes to their daily routine could allow them to improve their gut microbial diversity and health. P9 said, "I believe it's about adding

one more activity that's not necessarily taking away from my routine.” Relating to this, P7 said, “Now, I find myself replacing the tram ride with a short walk, Or walking on the grass barefoot instead of just walking around the yard (refers to the exposure to friendly microbes in soil and grass); having those extra steps, I think, all add up.”

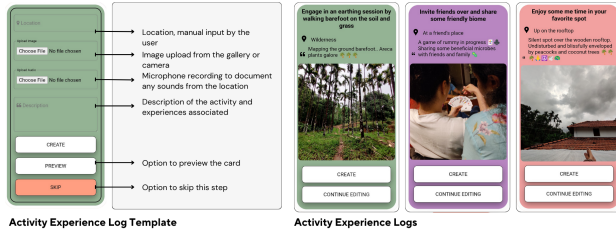


Figure 10: (Left) The anatomy of the player activity card logs (experience logs), which open after the completion of an activity. This step is optional in the game. (Right) Players have the option to preview the card they made.

5.2 Theme 2: Embodied Connections

This theme describes how the game fostered a personal connection between players and their gut biome through the game features and activity feedback. The findings suggest that participant engagement with the game evolved from focusing on in-game characters to a personal identification with their gut health, indicating a cognitive shift. However, discrepancies between the game’s feedback and players’ real-life efforts sometimes led to frustration, highlighting the challenges in accurately representing complex health dynamics in a game setting.

5.2.1 F6: Game features and feedback directed attention towards one’s friendly biome and its growth. Game features brought participants’ attention towards the friendly biome through synchronous and asynchronous feedback mechanisms. The Biome Energy Status is embedded with a synchronous feedback mechanism, directing participant attention to the needs of the friendly biome. For example, the visual depiction of low biome energy and population led participants to pay attention to it and act spontaneously to help it. P2 said, “Whenever I receive a low energy notification, I’d think, ‘Oh no, the bacteria! The balance must be off, and the bad biome might have increased. Let me do an activity!’” Witnessing the friendly biome increase on the home screen metre after an activity led participants to imagine and internalise similar growth of friendly biome in their gut. P4 said, “I imagined the friendly biome in my gut increasing and taking over, and I feel like I’ve contributed positively towards its improvement.”

Similarly, game features involving asynchronous feedback mechanisms, as in the slow growth of the friendly biome in the garden (Figure 9), directed participants’ attention to it through care, curiosity, and anticipation. P3 said, “The friendly biome felt like having pets that you need to care for.” Adding to this, P6 said, “I was curious to see the growth of new biomes that were generated through my activity.” About feelings of anticipation, P3 said, “Finding new biome friends in the garden the next morning was a feel-good moment. It

made me curious about the new biome friends I would make through my activity.”

5.2.2 F7: Game activities and their planning served as opportunities to connect with and increase one’s friendly biome. Engaging in-game activities involving self-care and mindfulness allowed participants to feel more connected with their biome. P4 said, “I felt like I could connect with my biome through self-care activities, such as meditation and sitting in silence.” About viewing weekly activities as opportunities to connect with one’s biome, P6 expressed, “I often have a schedule of the activities that I’m going to do during the week. Whenever I go through my schedule now, I think about the benefits these activities offer towards the growth of my friendly biome.” Similarly, P7 said, “Caring for my gut health has become an automatic part of my life. I now view some weekly activities as chances to increase my friendly microbes.”

5.2.3 F8: Interpretation of game concepts revealed cognitive shifts from player identity to self-identity. Interpretation of game activities surfaced shifts in how they were perceived across the study duration. Initial interpretation of the game activities was seen as benefiting the in-game character (e.g., Bugsy) without much thought on their interaction’s impact on their gut biome. P1 said, “The way that I interpreted the game activity engagement was less about my biome and more about the character, thinking that the more I engaged, the better off Buggy would be.” However, as their engagement continued, participants began to internalise the friendly biome in the game, thinking of it as their internal gut biome. P3 said, “I found the biome emojis in the flashcards particularly impactful as they shifted my focus from playing for my game character to realising that the game was ultimately for the benefit of my gut biome, making me more self-conscious and aware.” Playing the game reinforced the idea that their actions benefited their gut health. P8 said, “(...) it was clear to me that my actions and choices at play affect my gut biome and health.”

5.2.4 F9: Discrepancy in the game feedback versus real-life efforts led to frustration. Participants who were mindful of their daily diet recognised that the game’s representation of their gut may only be a rough approximation, causing them to be upset about the game’s inaccurate representation of the state of their gut. P6 said, “On days that I was conscious of my food intake and nutrition, I would get upset and frustrated if I saw the gut microbial balance was off or the biome unhappy and dull.” Furthermore, on such days, they were less focused on some game features and more focused on others. P6 added, “On days when the status of my gut was not accurately represented in the game, I was ignoring the microbial population bar on the home screen and instead turned my attention towards the movement of my friendly biome on the gut trail (map).”

5.3 Theme 3: Health Ownership

This theme describes how the unstructured nature of play in GGB allowed the freedom to engage with the game as they pleased, inspiring participants to manage their health activities creatively and independently. The game’s various activities motivated them to explore and engage in new, diverse health practices, instilling a sense of personal responsibility. This was further reinforced by flexible goal setting, where players set and pursued their health

targets, adopting a balanced approach to gut health engagement. Additionally, the game's Biome Connect flashcards and narrative messaging stimulated real-time reflection on health choices, encouraging players to be honest and reflective about their actions.

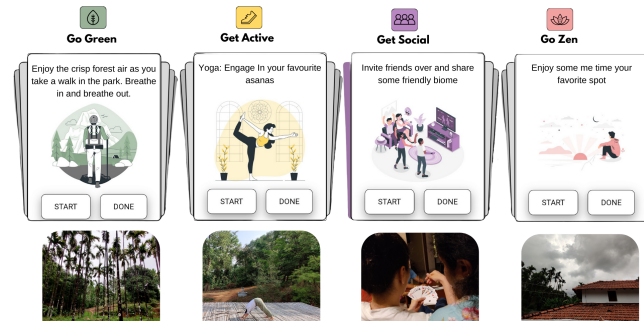


Figure 11: Four random cards from the four activity decks and the images captured during the activity.

5.3.1 F10: Variety and choice mechanics supported creativity and ownership of health decision-making. Participants appreciated the range of activities and the freedom GGB offers to choose their activities. This motivated them to explore ways to improve their gut health (Figure 11). P12 said, “The selection of the activities in the game motivated me to get outside and participate in diverse activities more frequently than I might otherwise.” The variety offered by the card decks provided ideas for creative planning of daily activities, allowing them to choose based on their energy level and activity intensity. P9 said, “On days that I’m recovering from my physical training, I could simply go for a walk (...) to keep myself moving and contribute to my balance.” Activity diversity even prompted engagement in activities outside their comfort zone. P12 said, “I’m not particularly inclined towards meditation or setting aside time for mindfulness. So, I tried to explore and engage in more of those kinds of activities.” Participation in-game activities led them to build a sense of accountability towards their gut health. P14 said, “Engaging in the game activities was about holding myself accountable.”

5.3.2 F11: Go-Go Biome supported flexible goal-setting. The game supported participants in pursuing their own goals. Overall, the goals varied between achieving a certain number and type of activities each week, achieving a certain balance in the biome population each day, or surpassing their previous day’s activity level. P4 said, “I challenged myself to increase the friendly biome population to at least 50% and above.” About beating past activity records, P12 said, “I aimed to surpass my previous records by focusing on activities I naturally enjoyed, rather than solely trying to achieve the game’s winning objective of 85:15.” About setting their own goals against the ultimate game objective and maintaining their own pace, P7 said, “Although there is an ideal game target of 85% friendly microbial growth, for some reason it didn’t bother me as much, maybe because I’m not competing with anyone. I am just competing with myself every day and trying to better it.”

5.3.3 F12: Biome Connect Flashcards led to momentary thinking about one’s actions in-situ. The Biome Connect flashcards (Figure 12) were intriguing and engaging as they provided timely information. P11 said, “The yes or no questions are interesting because that

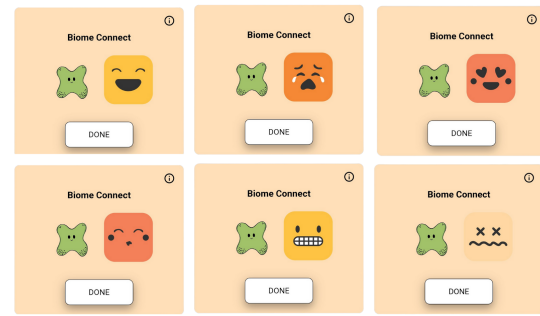


Figure 12: Go-Go Biome uses friendly biome narratives to evoke player empathy towards their friendly biome.

would tell me what does and does not help the friendly gut biome, (...) felt like “did you know?” facts motivating me to open the app and engage with it.” The appearance of the flashcards and the reactions from the biome through biome emojis prompted reflection in situ, causing them to think about their actions and decisions in real-time. P14 said, “There was a time when the flashcard regarding alcohol consumption popped up, just when I was having a beer with friends. I decided to be honest and answered ‘yes’. Some of me wished to have answered ‘no’ after watching the biome’s disappointment (Biome Emoji, See Figure 12). It was a moment of reflection for me.”

Similarly, P2 said, “Once, the biome connect card showed up when I was having a big bowl of udon noodles. Just then, the burger question (Would you like to feed the biome a greasy cheeseburger?) popped up. I remember it got me thinking about my bowl of noodles and how it may not be good for my gut biome.”

For greater effectiveness, suggestions were made to personalise the flashcard questions to help players adopt gut-friendly habits more readily. P12 said, “Including cards that make players think and adopt habits into their lives can be useful. For example, cards such as ‘Have you added iron-rich foods like spinach into your diet today?’ could potentially make me include them in my diet.”

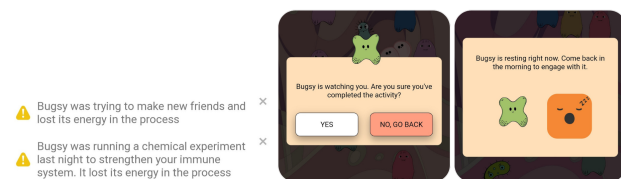


Figure 13: Go-Go Biome uses emojis and messages to support player conversations with the friendly biome character.

5.3.4 F13: Biome narrative and messaging facilitated self-introspection. Receiving a biome pop-up message whenever an activity is stopped led participants to reflect and be honest about their actions (Figure 13- centre). P8 said, “There was a time when I stopped an activity after a moment of starting it. A reminder popped up (“Bugsy is watching you. Are you sure you’ve completed the activity?”). This reminder felt

valuable because it made me reflect on being true to myself (...), reminding me that my activities affect me and no one else. It encouraged me to stay honest with myself even when no one's watching."

Regarding biome energy message prompts from the game (Figure 13- left), P2 said, "Receiving messages about the biome's energy level (Message: 'Bugsy was running a chemical experiment to strengthen your immune system. It lost its energy in the process.'). I could relate to the loss of the biome's energy, and it motivated me to engage in an activity to help it." Similarly (Figure 13- right), P8 said, "Trying to log an activity late in the night, I noticed the message ('Your Biome is resting right now, come back in the morning to engage with it.'). That felt like a gentle nudge (...) your biome is asleep, and so you must be too (...), reminding me to go to bed."

6 DESIGN LENSES FOR HEALTH ENGAGEMENT AND REFLECTION

Games for reflection have received tremendous attention in the last decade, which has led to a repository of design resources to design games and interactive technologies for reflection. This includes the articulation of types and levels of reflection [8, 28], designing personal informatics for reflection [4, 83], games for reflection [94], game design principles and reflective design strategies [45, 93]. Efforts have also been taken to understand and guide the evaluation of reflection-based designs [5]. These resources are generally useful in guiding design decisions and building game mechanics that employ resources such as temporality, conversational, comparison, and discovery to support health reflection. For example, our findings revealed that player experiences involving embodied connections were an outcome of using temporal, conversational, and discovery features (Figure 14). The experience of health ownership was seen as an outcome of using conversational and discovery features, and similarly, the outcome of incremental awareness was generated using comparison and discovery features. These findings are generally helpful in guiding future design decisions to craft specific user experiences using a combination of resources for reflection. However, our study findings further revealed the need to understand these design perspectives and resources through the lens of the 'biological self' or 'the body' to support health reflection. In doing so, it recognises the body as a team of interconnected actors and processes that work as a functional unit in the background, away from our immediate purview.

Our study findings (F6, F8) suggest that participants were reminded of the needs of their biome as separate from their own when interacting with some of the features of GGB. For example, some game reminders caused participants to think beyond themselves and focus instead on the friendly biome's struggles and needs in the game that mimics inner gut processes. This led them to engage spontaneously in an activity to help the friendly biome. Such separation from the self, if so, even momentarily, may be necessary to highlight and draw attention to the role of biological entities with whom we share background relationships, to bring us closer to them. These experiences closely align with our idea of considering the aspects of the biological body as a basis to support health reflection, therefore calling for an extension of the design perspectives by Bentvelzen et al. [8], namely Temporality, Conversation, Comparison, and Discovery. We do this by viewing them

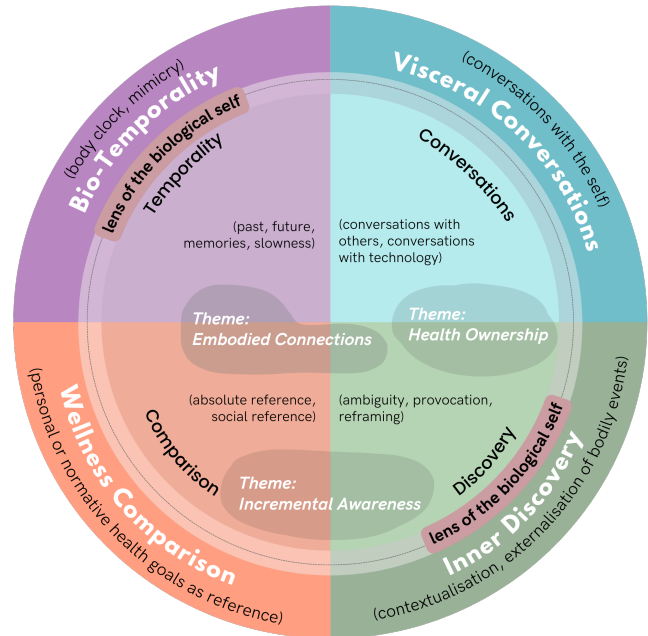


Figure 14: A mapping of the relationship between study findings, the current taxonomy, and its extension through the proposed design lenses for health engagement and reflection.

through the lens of the biological body and thereby propose four extensions in the form of lenses, namely *Bio-Temporality*, *Visceral Conversations*, *Wellness Comparison*, and *Inner Discovery* that serve as a starting point to understand and aid in the design for health reflection (Figure 14). These lenses extend the current taxonomy through the design perspectives (categories), each offering a design consideration (resource) that speaks to the lens. Overall, these four lenses serve as vantage points that can aid in the design for health reflection, and where the resources from the current taxonomy can support design researchers in achieving specific design goals.

6.1 Bio-Temporality (body clock- mimicry)

Bio-temporality as a design lens stemmed from participant experiences (F6, F7, F12), challenging the conventional understanding of time (i.e., in relation to oneself and the external world) and shifting attention to our inner biological self that sustains our external experiences and activities. This recognition further extends to the parts of our biological self that we share with more than human entities, like microorganisms, such as the gut microbiome and its ecosystem. The 'bio' in the bio-temporality lens recognises the internal biological actors, processes and needs to better align the external self to the inner biological self. The 'temporality' in bio-temporality relates to the concept of biological time or rhythm. Together, bio-temporality as a design lens is centred on bringing the user's attention to the natural cycles of their biological body and enabling them to be attuned to it while re-orienting their external life to support it. In relation to GGB, the focus is on bringing attention to the needs of the friendly gut microbiome and helping players reorient their habits and choices to support it.

Designing for bio-temporality requires understanding human biology's intricate rhythms, cycles, and temporal patterns, such as circadian rhythms, sleep-wake cycles, and metabolic processes. Such design recognises that individuals have distinct biological timeframes during which certain activities or health-related behaviours are more effective or beneficial. For instance, scheduling physical exercise during a person's peak energy levels or aligning meal timing with circadian rhythms for optimal metabolism. In doing so, bio-temporality differentiates itself from temporality, a more encompassing concept related to time and aspects of time. On the other hand, bio-temporality is a narrow and more specialised concept that pertains specifically to the alignment of health-related activities and interventions concerning human biological rhythms. Rapp et al. [85] suggest personal informatics should primarily focus on enhancing users' self-awareness and self-knowledge to promote reflection as a valuable goal in its own right. This deepened self-understanding can fundamentally change an individual's subjective experience, enriching it and eventually reconfiguring and reinterpreting their existing knowledge. Importantly, this process can likely result in behavioural changes. In the case of GGB, it offered a renewed understanding of the self in relation to the gut biome. As findings (F1, F2, F3, F7, F8, F10) suggest, the game offered opportunities for participants to reinterpret the game feedback through multiple perspectives (personal values, habits, other external factors, gut microbiome, and health). This concept closely relates to Sengers et al.'s [93] suggestion regarding designing features that offer interpretive flexibility, encouraging meaning-making through self-awareness.

Using synchronous and asynchronous feedback in GGB supported such interpretive flexibility. Findings (F6) highlight the role of synchronous and asynchronous game feedback in enhancing participants' understanding of the game concept, helping them relate to the needs of their friendly gut biome. For example, the synchronous feedback from changes in the Biome Energy Status supports immediate awareness of one's friendly biome, which symbolises the player's internal biological state. This awareness is crucial for bio-temporal understanding, as it allows players to make connections between their actions and the resulting influence on their gut microbiome. Alternatively, the asynchronous feedback, represented by the slow growth of the friendly biome in the garden sustains player engagement in game activities through a delayed game feedback mechanism. This mirrors the slower biological processes within the body, fostering a sense of anticipation and patience in players to appreciate the gradual nature of biological changes. These game features offer interpretive flexibility, encouraging meaning-making by allowing players to create a narrative around their gut health. Players attribute their actions to the growth of their friendly biome, forming a meaningful connection between their in-game activities and their internal biological processes.

6.1.1 Design game narratives and feedback that bring attention to one's body clock. Based on our findings (F6, F8), we suggest designers design narratives and mechanics that bring attention to one's body clock. Features that mimic inner body processes and rhythms can help users establish close connections with the inner body process. For example, in GGB, we designed momentary game

interactions with immediate and delayed feedback to sustain engagement. Prolonged interaction with the game interface is not the primary goal for games like GGB, which prioritises real-world participation as it can quickly lead to instances of disengagement [76]. Momentary interactions with immediate feedback in GGB are enabled through quick and easy logging after completing an activity, allowing participants to focus on supporting their biome's needs and witnessing the resulting changes in the game environment. Embedding momentary interactions, like quick tasks and simple app usage with frequent, low-complexity, and single-purpose actions such as tap gestures on widgets, can support immediate and delayed gratification [57].

Another method includes designing for momentary interaction coupled with delayed feedback, as seen in the growth of a new biome in Biome Garden. We also implemented time-based constraints on player interactions and time-delayed feedback (ambiguity), enhanced by friendly biome narratives (provocation), highlighting its needs for player relatability. Examples include indicating the friendly biome's need for nightly rest, thus restricting game interaction after 11 pm (narrative), gradually revealing the friendly biome in the biome garden (narrative & visualisation), and the game background's transition from dull to vibrant (visualisation). Such feedback can create less technologically demanding, engaging, and personally relevant game experiences. Previous studies have used slowness to foster empathy [32] and enhance understanding of the digestive process [48]. Designs representing inner body aspects can use slowness to mirror bodily changes throughout the day. Future developments in games like GGB could explore ambient gaming [98], using location and biofeedback technologies [72, 73] to focus on inner body processes and enable dynamic, prolonged engagement. This approach could let users naturally cultivate their microbial population over months, allowing experimentation with activities, establishing routines aligned with their biological clock, and fostering sustained engagement and reflection. As users become more invested in their health journey, they may find greater value in the experience.

6.2 Visceral Conversations: (*conversations with the self*)

The lens of visceral conversations is concerned with designing to support inner dialogue (conversations with the self) concerning one's values, practices, and behaviours that influence the biological aspects of the self (such as the digestive system or the autonomic nervous system). Visceral conversations as a lens extend the 'conversation' perspective by Bentvelzen et al. by turning the attention from conversations with others and technology to focusing on dialogic engagement with parts of the self that influence the inner biological self. Our findings (F3, F13) illustrate that features in GGB, such as logging of activity experiences and the Biome Connect, facilitated participant self-expression and introspection, fostering deep conversations with oneself regarding personal habits affecting their gut health (F4). Participants recognised and re-assessed their dietary and lifestyle choices, with some even seeking professional advice for improved nutritional balance (F5). Designing technology to stir inner dialogue about activities, values, and practices that affect various aspects of one's biological body and experience

aligns with the concept of visceral conversations. Here, technology acts as an impartial entity, posing questions without biases or preconceptions. This can encourage sincerity and honesty in one's thoughts and approach towards health actions, even when faced with discomfort from such interactions (F13). Khaled [45] suggests designing interactions that prompt users to prioritise questioning over simply finding answers, exploring alternative viewpoints, and developing their interpretations. Findings (F6, F9, F12) show that including positive and negative emojis in the biome garden and Biome Connect flashcards elicited a range of emotions, facilitating reflection on meal practices and daily activities. This approach acknowledges that critical reflection can sometimes be uncomfortable and may involve moments of frustration or uncertainty, aligning with the idea that disruption rather than comfort can leave a lasting impact (ibid). Research indicates that evoking negative emotions like frustration and disgust can effectively encourage attitude and behaviour changes [76], emphasising the importance of embracing a spectrum of emotions in design, rather than focusing solely on positive ones, to truly engage and foster reflection.

6.2.1 Design bite-sized science stories and hypothetical scenarios that question health assumptions and invite playful self-expression. Based on our findings (F12, F13), we suggest that designers use real-world examples and simple language to present intriguing facts and hypothetical scenarios, engaging scientific information. Familiar tools like emojis, sliders, and tap-to-find features can invite user attention. Findings (F8, F12) also suggest that daily self-reflection can occur through conversational features like biome connect flashcards, posing contextual questions to promote reflection in situ. Technology-mediated conversation via scenarios and questions has been used in games to encourage reflection on health behaviours. For instance, OrderUp [30] is a casual game that involves players in nutritional decision-making, where they recommend meals to various game personas and reflect on the nutritional value of these choices, fostering dialogic reflection from personal and alternative perspectives [28].

Our findings (F12) suggest that asking relevant, everyday questions about meals or health-related activities can prompt user thought. Direct questions like, “Did you consume your daily dose of fibre today?” or “Did you get your dose of sunshine today?” can encourage users to consider their current situation, leading to positive actions when opportunities arise. This approach aligns with experiential fidelity, where game scenarios prime users for real-life situations, acting as cues for reflection and action [39]. Designers can also consider including features for users to express their current state, like sleep, mood, or even the state of their gut, using simple UI elements and linking these to their gameplay for dynamic feedback. For instance, updating mood, meal diversity, and energy levels throughout the day via easy-to-use UI elements can serve as daily self-check-ins with oneself. Finch [27], a mental health self-care tool, uses a morning check-in where users tap an emoji to convey their feelings to their healthcare buddy. Features can be customised to make users empathetic to specific body organs and their tasks in keeping the user healthy. Such features can encourage visceral conversations, which may lead to self-reflection on personal habits that affect their work.

6.3 Wellness Comparison (*personal or normative health goals as reference*)

Wellness comparison is concerned with designing to support the comparison of one's current well-being to a desired state of wellness, where the goals for wellness are self-set or as suggested by the designed technology based upon pre-set or personalised health norms. Our findings (F10, F11) suggest that the choice of activities in the decks and the biome population metre enabled flexible goal-setting based on individual preferences and objectives. Some participants focussed on achieving specific activity milestones, while others prioritised balancing the biome population or surpassing their past activity records. This aspect aligns with wellness comparison, where personal goals and progress are measured against personal or normative standards. Similarly, finding (F3) highlights the importance of features like Biome Digest and Activity History in helping participants reassess and compensate for missed activities. These features helped them reflect on past activities and plan according to personal priorities, encouraging proactive management of wellness strategies. This suggests the design's role in supporting users in reflecting on their lives, addressing a core principle in reflective design [93]. Findings (F10) indicate that activity variety helped participants identify patterns in their choices, surfacing the need to diversify activities for balanced gut health. These findings mirror the wellness comparison concept, where individuals assess their current behaviours and consider adjustments to achieve a balanced and holistic approach to wellness.

6.3.1 Design to encourage personal goal setting in the presence of a standardised health norm as a reference. Based on our findings (F11), we suggest designers consider allowing users to set their own goals against a normative ideal goal provided by the system based on scientific health recommendations. Doing so can help cater to different player types – the ones that need some handholding and motivation and those that are intrinsically motivated and have a clear idea of what they want to do for their health. For instance, FoodInTake Logger by [42] allows users to manually track vegetable intake and shows how many more servings are needed daily. This visual feedback helps users reflect on their intake compared to recommended levels, potentially prompting behavioural change. The absolute reference serves as a comparative tool and a trigger for reflection.

Findings (F10, F11, F3) highlighted game objectives and activity diversity, empowering participants to fix their goals and plan based on their needs and priorities and fostering accountability and control over their health activities. In solo engagement, designers can consider inviting users to beat and exceed their records by displaying past accomplishments alongside unexplored activities and challenges to inspire action. Additionally, this can be achieved by offering diverse options: 1) to choose health activity within and across categories (environment, physical, social, mindfulness); 2) to choose activity intensity in case of physical activities; and 3) to explore various activity locations. This can be further enhanced by encouraging user-defined activities for personalisation [107]. As revealed in our findings, apart from enabling rich experiences of their health activity, the freedom enabled through game mechanics and game activity diversity can support self-directed action.

6.4 Inner Discovery (*contextualisation, externalisation of bodily events*)

Inner discovery is concerned with designing to support an embodied sense of self-awareness of our biological self. This lens builds on and extends the ‘discovery’ perspective by redirecting users’ attention towards lived inner body experiences and their awareness of them through external events. It involves a deeper understanding of one’s body and the recognition of internal aspects like the gut microbiome, even if so at an abstract level, within the context of personal well-being. This lens shares close links with the lens of bio-temporality, which is concerned with bringing attention to one’s body clock and rhythm. While it can even be argued that the previously described lenses (*bio-temporality*, *visceral conversations*, and *wellness comparison*) may contribute to inner discovery in some way, the lens of inner discovery is more encompassing. Central to the lens of inner discovery is the phenomenological experiences of inner body events supported through playful discovery, contributing to an embodied awareness of the self, enhancing one’s connection, and understanding of one’s body and functions.

Findings (F6, F7) revealed that participants embodied the game feedback of increasing the friendly gut biome and decreasing the unfriendly biome whenever they engaged in an activity, causing them to imagine its occurrence within their gut, even outside the gameplay (F6, F2). This suggests that engagement in-game activities primed participants to reimagine and relieve the potential effects of their everyday activities on the friendly biome. Such embodied awareness of one’s inner body processes aligns with the inner discovery lens, which encourages individuals to think more deeply about their actions and the related consequences on their biological processes. We highlight two conceptual resources implemented in GGB that supported participants’ inner discovery: contextual links and cues and externalisation.

Firstly, with GGB, we wanted to extend players’ understanding of gut health beyond the lens of diet, providing an extended contextual lens through which they understand gut health i.e., the wider factors and the real-world scenarios that affect it. Spending time in nature, leading a physically active life, eating well-balanced food, or engaging in mindfulness activities are popularly known to benefit our health. Findings (F1, F2, F7) revealed that game interactions and activities led to shifts in mindset, reframing these known activities and causing some to re-evaluate personal decisions surrounding gut health. Such reframing allowed participants to view their daily actions from an inner biological perspective, encouraging sense-making of the impact of their routine activities on their gut biome and health. This led participants to recognise real-world cues in these activity contexts, prompting further activity engagement and reflection (F2). Research suggests that context is unobserved and needs to be inferred [33]. Viewing these activities through the lens of the gut microbiome offered a new perspective to understand the missing link of why these activities are beneficial to one’s health. Participants rediscovered new perspectives based on familiar actions, bridging the known and unknown [93].

Secondly, GGB externalised complex inner processes like the growth and decline of friendly and unfriendly biome, biome energy levels, and the introduction of new friendly biome characters, enhancing player engagement through playful visualisation and

empathetic narratives (F6, F13). Central to this was the character Buggy, portrayed with human-like emotions through emojis in the biome garden, representing different energy states. These emotions and auditory feedback were reinforced by game narratives, reminders, and pop-up messages, effectively communicating Buggy’s emotional state and needs, such as low energy or threats from unfriendly biomes. This approach encouraged players to connect with Buggy empathetically and actively participate in the game by responding to its needs, as shown by their eagerness to assist Buggy when notified (F6, F13). The anthropomorphic (biome characters and emoji) and metaphoric (game environment changes) representation deepened the understanding of the relationship between player actions and game feedback, which aligns with prior findings from works like WhoisZuki and UbiFit Garden [21]. Fostering embodied self-awareness of biological processes through externalisation and involving players in unravelling links between game events and internal biological phenomena aligns with the concept of inner discovery.

6.4.1 Embed contextual cues and links in the game mechanics to support embodied health awareness. Designers can consider embedding contextual cues in-game activity cards, linking game insights with real-world contexts through visuals and text. Findings (F6, F7) revealed that repetitive interaction with the game scenarios and feedback primed participants to recognise these contexts and health behaviours (F2), learning to infer these cues whenever the context presented itself. Pasumarthy et al. [79] explore the use of contextual cues from a dining setting to support an understanding of gut health from a dietary perspective. Their findings revealed that repetitive interaction with dietary aspects served as cues, reminding participants of the benefits of healthy meal practices outside the game, thus blurring the boundaries of play. Future designs could use location-based gaming involving various contexts to enhance gut health engagement and reflection, where different spatial regions and activities can uncover different experiences and potentials for understanding our connection with these spaces through the microbiome that we share and exchange. This also opens opportunities for playful HFI by exploring play potentials with food and spaces that grow food, like local community farms. Pervasive games can help support such play experiences by adding a digital layer to real-world contexts [55], making the complex aspects of gut health, like microbial activities and digestive processes, more accessible and engaging. These games can help turn everyday objects and spaces into exploratory tools for gut health, revealing the microscopic layers crucial to our bodily functions and contributing to embodied self-awareness of health processes.

7 LIMITATIONS AND FUTURE WORK

This research explored the design and evaluation of the casual smartphone game Go-Go Biome, focusing on engagement and reflection on gut health. We recognise some limitations, including the small sample size considering the complex nature of gut health that is influenced by various external and internal factors, often leading to diverse individual experiences. However, the game serves as an initial step in linking familiar activities to gut health, stimulating mental connections and emphasising the pivotal role of gut health in overall well-being [105]. Secondly, excluding dietary

habits from the game caused some participants to feel disconnected from their gut health representation. Future research can consider incorporating diet and sleep factors to provide a more comprehensive understanding of daily activities and their impact on gut health, encouraging reflection. Finally, we acknowledge the shorter duration of this study, considering that the game focuses on engagement in real-world activities. Our exploratory research is only an initial step towards such understanding. Future design explorations can consider how participants' attitudes and behaviours evolve over more extended periods to determine the long-term impact of the game on their gut health awareness and routines.

8 CONCLUSION

In conclusion, our research highlights the critical need for a holistic approach towards understanding and supporting gut health engagement, involving diverse perspectives such as diet, physical activity, lifestyle, and environment. The design and analysis of the Go-Go Biome game offer rich insight into the design of interactive technologies to support users in reflecting on everyday choices and activities that may affect gut health. By employing interactive visuals, unstructured play mechanics, and reflective design principles, the game not only visualises the interplay between beneficial and harmful gut microbes but also encourages actions that may promote a healthy gut microbiome.

A field study of Go-Go Biome with 14 participants, validated the game's effectiveness in boosting awareness, engagement, and reflection on gut health factors. Central to our study findings was the need to incorporate the 'biological self' in the design of interactive technologies that aim to foster health reflection and engagement, thus inspiring the proposal of four design lenses—*bio-temporality*, *visceral conversations*, *wellness comparison*, and *inner discovery* – to enrich future design explorations.

As we look ahead, there is a promising avenue for future explorations using physical ambient technologies to support gut health engagement and reflection in personal and shared settings. Physical interfaces can be designed to serve as unobtrusive ambient systems, by embedding subtle, health-promoting interactions into everyday life. By minimising direct interaction and maximising intuitive support, ambient interfaces can significantly enhance gut health understanding and engagement. This approach not only aligns with the need for holistic health strategies but also uncovers new possibilities for designing environments that encourage healthier choices in both personal and communal spaces, thus contributing to a broader understanding and appreciation of gut health's role in overall human health. As we continue to navigate the complexities of gut health, the exploration of innovative technologies and design paradigms will be instrumental in creating more informed, health-conscious societies.

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